Object Detection Extension

https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner

Python | PyTorch | ONNX Runtime | RapidMiner | Object Detection

OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

IMPACT

Developed two RapidMiner operators for object detection — FineTuneObjectDetectionModel and ObjectDetectionModelInference — enabling users to train and deploy detection models within a visual workflow without writing code.

TECHNICAL WALKTHROUGH

Object Detection Extension

Problem Statement
Different object detection use cases require different trade-offs between accuracy, speed, and computational cost. A single model cannot satisfy all scenarios.

Approach
Developed a unified object detection extension using PyTorch, supporting

multiple architectures

- Faster R-CNN
- FCOS
- SSD
- SSD Lite
- RetinaNet
- YOLO (extended work)
- Training
- Dataset annotated using XML bounding boxes

- Transfer learning using COCO pre-trained weights

Fully configurable training pipeline

- Model selection
- Batch size
- Epochs
- Learning rate
- Momentum
- Weight decay

Evaluation metrics

- IoU (Intersection over Union)
- Precision
- Recall (per class)
- Inference
- Users can choose any trained or pre-trained model

Output includes

- Image with bounding boxes, labels, and confidence scores
- Tabular output with detected objects
- Results also generated in base64 format for easy deployment and
- integration
- Key Observations

- Faster R-CNN performed best for accuracy
- SSD and SSD Lite were faster but had lower recall
- RetinaNet handled class imbalance well
- Model choice depends on application requirements

KEY DETAILS

- FineTuneObjectDetectionModel accepts image and annotation directories (YOLO/Pascal VOC) and outputs model weights and class list CSV.
- ObjectDetectionModelInference runs predictions on new images and returns bounding boxes and class labels as a DataFrame.
- Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.
- Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner Studio.
- Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process designer.